

Oswald Gyabaah

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PROFESSIONAL SUMMARY

Full-stack engineer with 6 years shipping production JavaScript and TypeScript across React/Next.js, Node.js, and Python. Founded and scaled two platforms from zero - MyHealthCop (healthcare, 3 applications on AWS EKS/ECS with Lambda for event-driven workloads) and Virk Cloud (a multi-tenant Kubernetes Platform-as-a-Service) - and currently own a national bank's core banking cluster sustaining 1.5M+ daily transactions with 99.97% uptime. Comfortable inheriting unfamiliar production systems: led the GTBank Basis-to-Finacle core banking migration cutover across 33 applications in two weeks. Strong in AWS serverless and container stacks (Lambda, EventBridge, DynamoDB, EKS, ECS, S3, SQS/SNS, VPC, IAM), Terraform IaC, and NoSQL data modeling across DynamoDB, MongoDB, Firestore, and Convex with both single-table and multi-table design patterns. --- - Full-stack JavaScript and TypeScript - React and Next.js production applications - Node.js (Express) and Python (FastAPI, Django) backends - AWS Lambda, EventBridge, and DynamoDB - NoSQL data modelling: DynamoDB (single-table and multi-table), MongoDB, Firestore - AWS infrastructure and Terraform IaC (CloudFormation-adjacent) - Inheriting and evolving live production codebases - Mission-driven product delivery ---

CERTIFICATIONS

AI Engineering Specialization (ByteByteAI), Mar 2026: Agents, RAG, LLM applications

Oracle Certified DevOps Professional (Oracle), Aug 2025

AWS Certified Cloud Practitioner (AWS), May 2024

Database Structures and Management with MySQL (Coursera), Dec 2022

Django Web Framework (Coursera), Dec 2022

SKILLS

Frontend: JavaScript, TypeScript, React.js, Next.js, Tailwind CSS

Mobile: Flutter (iOS + Android), Codemagic CI/CD

Backend: Node.js (Express), Python (FastAPI, Django, Celery), REST APIs, GraphQL, WebSockets

Databases (NoSQL): DynamoDB (single-table and multi-table designs), MongoDB, Firebase/Firestore, Convex, Redis

Databases (SQL): PostgreSQL, MySQL, SQLite

AWS: Lambda, EventBridge, DynamoDB, EKS, ECS, S3, SQS/SNS, VPC, IAM

Other cloud: Azure, GCP (GKE, Cloud SQL, Cloud Storage), OCI

IaC & CI/CD: Terraform, Ansible, GitHub Actions, FluxCD, ArgoCD, Azure DevOps, Codemagic, Jenkins

Containers & Orchestration: Docker, Docker Compose, Kubernetes (EKS, GKE, TKG, RKE2), Helm

Messaging & Queues: AWS SQS/SNS, Kafka, RabbitMQ, Redis queues

Observability: Prometheus, Grafana, ELK Stack, Sentry

AI & Agents: LLM API integration (Claude, OpenAI), RAG systems, agent development, tool-use patterns

WORK EXPERIENCE

Guaranty Trust Bank Ghana LTD

Apr 2024 - Present

Accra, Ghana

Team Lead, Core Banking Services

May 2025 - Present

- Built a bank-wide OTP service using Django (Python) and Next.js (React/TypeScript), integrating SMS and email gateways, to secure core banking access for 600+ internal users with structured logging to Elasticsearch via Filebeat.
- Built a national ID verification service using Django and Next.js, processing and storing ID documents in self-hosted MinIO with automated verification and customer record updates.
- Led a 5-person QA team through the bank's Basis-to-Finacle core banking migration cutover, designing and executing 10,000+ test cases across 13 core banking modules and 33 internal applications over a two-week window, covering integration, UAT, and peripheral device validation.
- Own the on-prem Finacle core banking deployment on Tanzu Kubernetes Grid, managing deployments, patching, and incident response across 800+ production pods to sustain 1.5M+ daily transactions.
- Led L2 incident response, correlating logs across internal apps and core banking microservices via the Elastic Stack for root-cause analysis and remediation.

DevOps Engineering Lead

Apr 2024 - Present

- Founded the bank's DevOps team of 4 engineers, replacing manual code handoffs with enforced Git workflows, containerized Kubernetes deployments, and automated release pipelines.
- Implemented CI/CD automation that cut deployment time by ~90% and increased release velocity ~5x; introduced GitOps with FluxCD for consistent, repeatable deployments; integrated Veracode and Trivy scanning for automated security analysis.
- Architected high-availability Kubernetes clusters across multiple environments running internal applications handling 3M+ daily requests, with 99.97% measured uptime inclusive of monthly scheduled maintenance.
- Managed multi-cloud infrastructure across AWS and Azure: resource provisioning, private VPC peering to on-prem systems, IAM, access control, and cost optimization.
- Automated 95% of infrastructure configuration management using Terraform, Ansible, Python, and Bash, turning multi-day manual processes into reproducible, version-controlled deployments.

MYHEALTHCOP LTD

Jan 2021 - Mar 2024

Chief Technical Officer (Founding)

Accra, Ghana

- Architected the myhealthcop.com platform end-to-end across requirements, system design, database modeling, and API development, delivering three applications: a patient-facing Flutter mobile app (iOS + Android), a Flutter app for health professionals, and a React admin web dashboard.
- Designed secure, cost-efficient AWS infrastructure running containerized microservices on EKS and ECS alongside Lambda functions for event-driven workloads, with autoscaling, load balancing, and inter-service communication.
- Built a high-throughput background processing system handling 1M+ daily jobs across payments, messaging, and notifications using SQS/SNS, with SMS, email, and payment provider integrations including multi-provider failover.
- Led performance optimizations that reduced peak request latency by 90% through Redis caching, query tuning, and indexing in PostgreSQL.
- Automated application delivery and infrastructure provisioning via CI/CD pipelines on GitHub Actions and Terraform, making deployments repeatable, auditable, and reversible.
- Led a 12-person cross-functional team (frontend, backend, mobile, product, design, QA) through full project lifecycle under Agile/Scrum.

PROJECTS

Virk Cloud (virk.cloud)

Aug 2025 - Present

Founder & CTO

- Architected a multi-tenant Platform-as-a-Service for startups with built-in CI/CD, fixed billing, observability, real-time log streaming, and alerting.
- Built the full management console in Next.js (React, TypeScript) with authentication, user management, real-time monitoring dashboards, and billing - inherited-and-evolved an early prototype into production.
- Manage the underlying infrastructure on virtual private servers: RKE2 cluster deployment, hardware/storage provisioning, network and traffic management, and DNS.

Stack: Next.js, React, TypeScript, Convex, Tailwind CSS, Kubernetes (RKE2), Terraform, FluxCD, Prometheus, Grafana, Python

Digital-Recon: Transaction Reconciliation Engine

2025 - Present

Founder

- Building a real-time transaction reconciliation engine for financial institutions, detecting discrepancies across heterogeneous data sources including relational DBs, REST APIs, Elasticsearch, and Kafka topics.
- Designed a Next.js/TypeScript triage dashboard enabling operators to define reconciliation rules and resolve mismatches with a full audit trail.
- Architected an event-driven processing layer using Kafka for upstream ingestion and Celery for scheduled and on-demand reconciliation jobs.

Stack: React, TypeScript, Next.js, Python, Celery, Kafka, Elasticsearch, PostgreSQL, Docker

Education Research and Innovation Foundry (erifoundry.org)

Oct 2025 - Present

Founder & Research Lead

- Leading design of AI-driven research initiatives in adaptive learning, cognitive modeling, and educational analytics.
- Coordinating cross-disciplinary research on ML and software tooling for education.
- Prototyping applications of LLMs in human-centered learning systems.

AI Agents for Infrastructure Operations

2024 - Present

Personal R&D

- Building specialized AI agents for QA automation, code review, and sysadmin operations, running against live Linux production servers with real workloads.
- Built AWS Lambda functions triggered by EventBridge to deliver notifications on codebase events.
- Used DynamoDB for NoSQL data modeling across personal projects, with both single-table and multi-table design patterns.

EDUCATION

Ashesi University | BSc Computer Engineering | Sep 2016 - May 2020 | Berekuso, Ghana